

ENVIRONMENTAL SCIENCE

Maximum Marks: 80

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during the first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. *The time given at the head of this Paper is the time allowed for writing the answers.*
-
5. *Attempt **all** questions from **Section A** and **any four** questions from **Section B**.*
 6. *The intended marks for questions or parts of questions are given in brackets [].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 9 printed pages.

SECTION A (40 Marks)

(Attempt *all* questions from this *Section*.)

Question 1

[16]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write the correct answers only.)

- (i) Which of the following is a disadvantage of using CNG?
- (a) Eco-friendly
 - (b) Cheaper than fossil fuels
 - (c) Gives less mileage
 - (d) Low maintenance
- (ii) **Assertion (A):** Education plays a fundamental role in empowering women.
Reason (R): Access to education misguides women to make informed choices about their lives and future.
- (a) (A) is true and (R) is false.
 - (b) (A) is false and (R) is true.
 - (c) Both (A) and (R) are true and (A) is the correct explanation of (R).
 - (d) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
- (iii) A city designed to balance economic growth, environmental protection, and social well-being is referred to as a:
- (a) Secondary city
 - (b) Sustainable city
 - (c) Smart city
 - (d) Sprawling city

- (iv) Match Column 1 with Column 2 and choose the correct option from those given below.

COLUMN 1

1. National Park
2. Wildlife Sanctuaries
3. Biosphere Reserves
4. Small dams

COLUMN 2

- A. Check Dams
- B. Great Nicobar Island
- C. Gir
- D. Kanha

- (a) 1-C, 2-D, 3-A, 4-B
- (b) 1-D, 2-C, 3-B, 4-A
- (c) 1-A, 2-C, 3-B, 4-D
- (d) 1-B, 2-A, 3-C, 4-D

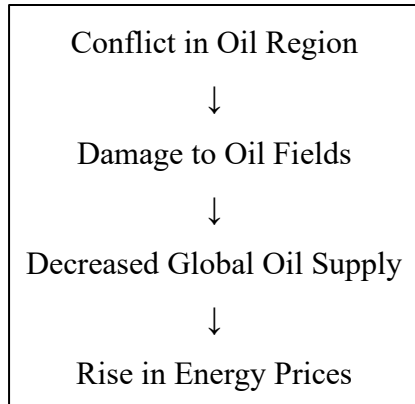
- (v) Which of the following is a problem that has emerged with the overuse of pesticides?

- (a) Better crop yield
- (b) Soil aeration
- (c) Pesticide-resistant pests
- (d) Lesser crop failure

- (vi) An area for conservation of species that may not survive in the wild.

- (a) Wildlife Sanctuary
- (b) Captive Breeding
- (c) Biosphere Reserve
- (d) National Park

- (vii) Which sector will be affected **first** by rising energy prices as shown in the flowchart given below?



- (a) Transportation sector
 - (b) Education sector
 - (c) Tourism sector
 - (d) Banking sector
- (viii) Waste such as old or broken CPUs, laptops, printers, etc. can be categorised as:
- (a) Industrial waste
 - (b) Agricultural waste
 - (c) Domestic waste
 - (d) Electronic waste
- (ix) India has operational branches of Amazon, hence it is called a:
- (a) Home country
 - (b) Host country
 - (c) Developing country
 - (d) Third-world country

- (x) 1. GIS helps identify the latitudinal and longitudinal location of an object on the Earth.
2. GPS is only used to determine the location of stationary objects.
3. GIS technology is widely used in vehicles, planes and ships.
- (a) Only 1 is correct.
- (b) Only 1 and 3 are correct.
- (c) Only 2 and 3 are correct.
- (d) All statements are correct.
- (xi) Which of the following air pollution control devices has maximum efficiency?
- (a) Cyclone separators
- (b) Fabric filters
- (c) Wet scrubbers
- (d) Electrostatic precipitators
- (xii) Which of the following is **NOT** an effect of the population explosion?
- (a) Increase in the demand for food
- (b) Decrease in infant mortality
- (c) Depletion of natural resources
- (d) Increase in waste generation

- (xiii) To make the construction activities more environmentally friendly, which of the following should **NOT** be undertaken?
- (a) Reducing construction waste
 - (b) Using eco-friendly construction material
 - (c) Using timber for construction
 - (d) Efficiently using water resources
- (xiv) Narmada Bachao Andolan : Medha Patkar : : Sukhomorji : _____
- (a) P.R.Mishra
 - (b) Aamir Khan
 - (c) Anna Hazare
 - (d) Amrita Devi
- (xv) When we read about Asian Elephant, Red Panda, Blue Whale, we are referring to:
- (a) Endangered species
 - (b) Endemic species
 - (c) Vulnerable species
 - (d) Extinct species
- (xvi) Identify the incorrect pair:
- (a) CITES – 1975
 - (b) IUCN – 1948
 - (c) Project Tiger – 1974
 - (d) Ramsar Convention – 1971

Question 2

- (i) Name *two* liquid fuels from biomass. [2]
- (ii) How has economic development in developing countries affected the environment? [2]
- (iii) State *two* goals of using alternative technology to create a self-sustaining society in India. [2]
- (iv) How do landfills pose a challenge to the environment? [2]

Question 3

- (i) Name *two* energy efficient devices to control air pollution from industries. [2]
- (ii) How does a growing population result in a climate crisis? [2]
- (iii) Urbanisation contributes to economic growth and improved standard of living, yet it poses a challenge to the future.
State *two* drawbacks of urbanisation. [2]
- (iv) State *two* objectives of Integrated Rural Development (IRD). [2]

Question 4

- (i) How does Polyvarietal cultivation differ from polyculture? [2]
- (ii) What does IUCN stand for? [2]
- (iii) What is biomass energy? [2]
- (iv) What is composting? [2]

SECTION B (40 Marks)

(Attempt **any four** questions from this **Section**.)

Question 5

- (i) (a) What are *MNCs*? [5]
- (b) State *three* immediate impacts of the Bhopal Gas Tragedy.
- (ii) (a) Name the treaty designed to protect the Earth's ozone layer. [5]
- (b) Explain the importance of the ozone layer and how the ozone layer gets depleted.

Question 6

- (i) (a) What are *catalytic converters*? [5]
- (b) State *three* economic policy measures to control vehicular pollution.
- (ii) (a) Name the campaign in India to promote girls' education. [5]
- (b) Explain *two* economic development and two women-centred development measures to control population growth.

Question 7

- (i) (a) What are *secondary cities*? Name a secondary city in Maharashtra. [5]
- (b) State *three* advantages of developing secondary cities.
- (ii) (a) What is *reforestation*? [5]
- (b) State *two* consequences of overgrazing and *two* provisions included under the National Forest Policy, 1988.

Question 8

- (i) (a) What is *trickle drip irrigation*? [5]
- (b) State *two* advantages and *two* disadvantages of trickle drip irrigation.
- (ii) (a) What is *Ex-situ conservation*? [5]
- (b) What role do *Seed Banks* and *Gene Banks* play in ex-situ conservation?

Question 9

- (i) (a) What is *nuclear fission*? [5]
- (b) List *four* disadvantages of nuclear energy.
- (ii) (a) What is *non-biodegradable waste*? [5]
- (b) State *two* ways in which non-biodegradable waste causes environmental pollution. Name *one* method of managing non-biodegradable waste.

Question 10

- (i) (a) What are *GMOs*? [5]
- (b) State *two* advantages and *two* challenges associated with GMOs.
- (ii) (a) What is *satellite imagery*? [5]
- (b) State *four* advantages of satellite remote sensing.